

1. Nov/2023/Paper_9701/41/No.2

(a) Define K_w mathematically by completing the expression.

$K_w =$ [1]

(b) Two solutions, **V** and **W**, are described.

- **V** is HCl(aq) .
- **W** is NaOH(aq) .
- The concentration of HCl in **V** is the same as the concentration of NaOH in **W**.
- The pH values of **V** and **W** differ by exactly 11.00 at 298 K.

(i) Calculate the concentration of HCl in **V**.

concentration of HCl in **V** = mol dm^{-3} [2]

(ii) Equal volumes of the two solutions **V** and **W** are mixed, giving solution **X**.

Name solution **X** and state its pH.

solution **X** pH

[1]

(iii) A 1 cm^3 sample of 1.0 mol dm^{-3} HNO_3 is added to 100 cm^3 of solution **X**, forming mixture **Y**.

A 1 cm^3 sample of 1.0 mol dm^{-3} KOH is added to 100 cm^3 of solution **X**, forming mixture **Z**.

Estimate the pH of mixtures **Y** and **Z**. No calculations are required.

mixture **Y** mixture **Z**

[1]

(c) (i) $\text{CH}_3\text{CH}_2\text{COOH}$, $\text{CH}_3\text{CCl}_2\text{COOH}$ and H_2SO_4 are all acidic.

Suggest the trend in the relative acid strength of these three compounds.

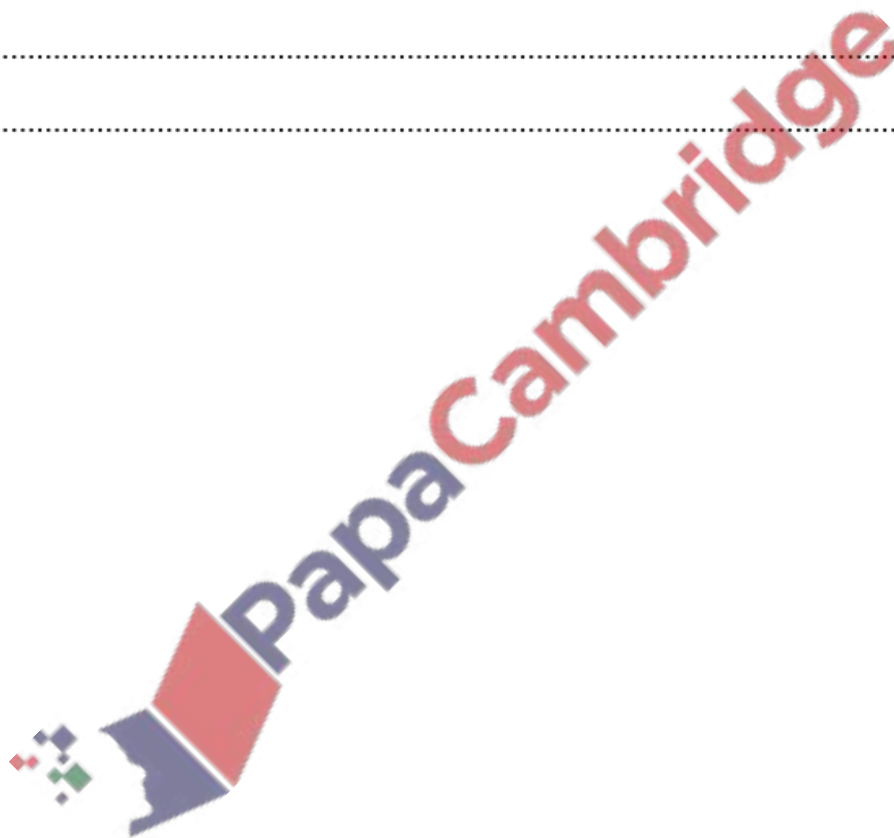
Explain your answer.

.....
strongest acid weakest acid

explanation

.....
.....
.....
.....

[3]



(ii) When concentrated H_2SO_4 is added to water a series of acid-base reactions occurs.

There are three conjugate acid-base pairs that can be identified during this series of reactions.

Write the formulae of these three conjugate acid-base pairs.

conjugate acid 1 conjugate base 1

conjugate acid 2 conjugate base 2

conjugate acid 3 conjugate base 3

[2]

(d) The partition coefficient, K_{pc} , of a substance, **Q**, between hexane and water is 7.84 at 298 K.

Q is more soluble in hexane than it is in water.

(i) Define partition coefficient, K_{pc} .

.....
..... [1]

(ii) 5.00 g of **Q** is shaken with a mixture of 100.0 cm³ of water and 100.0 cm³ of hexane at 298 K and left until there is no further change in concentrations.

Calculate the mass of **Q** dissolved in the water.

mass of **Q** = g [1]

(iii) A sample of **Q** is shaken with a different mixture of water and hexane and left until there is no further change in concentrations.

It is found that the mass of **Q** dissolved in each solvent is the same.

Use the K_{pc} value to suggest possible values for the volume of water used and the volume of hexane used.

volume of water = cm³

volume of hexane = cm³
[1]

(iv) **Q** is more soluble in hexane than it is in water.

It is suggested that **Q** is one of KCl , $\text{CH}_3(\text{CH}_2)_4\text{OH}$ or HCOOH .

Identify **Q**. Explain your answer.

.....
..... [1]

[Total: 14]

Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$, is a weak acid. The K_a of benzoic acid is $6.31 \times 10^{-5} \text{ mol dm}^{-3}$ at 298 K.

A 1.00 dm^3 buffer solution is made at 298 K containing 1.00 g of $\text{C}_6\text{H}_5\text{COOH}$ and a slightly greater mass of sodium benzoate, $\text{C}_6\text{H}_5\text{COO}^-\text{Na}^+$.

This buffer solution has a pH of 4.15.

(a) Define buffer solution.

.....
 [1]

(b) Write equations to show how this solution acts as a buffer solution when the named substances are added to it:

(i) dilute aqueous sodium hydroxide

..... [1]

(ii) dilute aqueous nitric acid.

..... [1]

(c) Calculate the H^+ concentration and the $\text{C}_6\text{H}_5\text{COOH}$ concentration in the buffer solution described. Use the expression for the K_a of $\text{C}_6\text{H}_5\text{COOH}$ to calculate the concentration of $\text{C}_6\text{H}_5\text{COO}^-\text{Na}^+$ in the buffer solution.

Show your working and give each answer to a minimum of **three** significant figures.



$$[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$$

$$[\text{C}_6\text{H}_5\text{COOH}] = \dots\dots\dots \text{mol dm}^{-3}$$

$$[\text{C}_6\text{H}_5\text{COO}^-\text{Na}^+] = \dots\dots\dots \text{mol dm}^{-3}$$

[3]

- (d) A 10.0 cm^3 sample of the buffer solution is mixed with 10.0 cm^3 of 1.00 mol dm^{-3} KOH. Both solutions are at 298 K. A reaction is allowed to occur without stirring.

Two observations are recorded:

- the temperature, after the reaction is complete, is fractionally above 298 K
- the pH, after the reaction, is greater than 13.

Explain these two observations.

.....

 [2]

- (e) Magnesium benzoate, $\text{Mg}(\text{C}_6\text{H}_5\text{COO})_2$, has a solubility in water of less than 1.00 g dm^{-3} at 298 K.

$$K_{\text{sp}} = [\text{Mg}^{2+}][\text{C}_6\text{H}_5\text{COO}^-]^2 = 1.76 \times 10^{-7} \text{ at } 298 \text{ K}$$

- (i) Calculate the solubility of $\text{Mg}(\text{C}_6\text{H}_5\text{COO})_2$ in water at 298 K. Give your answer in g dm^{-3} .

Show your working.

$[M_r: \text{Mg}(\text{C}_6\text{H}_5\text{COO})_2, 266.3]$

solubility = g dm^{-3} [2]

- (ii) An excess of $\text{Mg}(\text{C}_6\text{H}_5\text{COO})_2$ is added to a sample of 0.50 mol dm^{-3} MgSO_4 at 298 K.

State whether the equilibrium concentration of $\text{Mg}(\text{C}_6\text{H}_5\text{COO})_2$ is higher than, the same as, or lower than your answer to (i). Explain your answer.

The concentration is the concentration in (i).

explanation

..... [1]

[Total: 11]

3. June/2023/Paper_9701/41/No.5(d, e)

(d) Isocyanic acid, HNCO , can form cyanuric acid, $\text{C}_3\text{H}_3\text{N}_3\text{O}_3$, under certain conditions.

$\text{C}_3\text{H}_3\text{N}_3\text{O}_3$ has a cyclic structure containing alternating carbon and nitrogen atoms in the ring system.

Suggest a structure for cyanuric acid.

[1]

(e) Isocyanic acid, HNCO , is a weak acid.



(i) Write the mathematical expressions for $\text{p}K_{\text{a}}$ and pH .

$\text{p}K_{\text{a}} = \dots\dots\dots$

$\text{pH} = \dots\dots\dots$

[1]

(ii) Calculate the pH of $0.120 \text{ mol dm}^{-3}$ $\text{HNCO}(\text{aq})$.
Give your answer to **three** significant figures.

$\text{pH} = \dots\dots\dots$ [2]

(iii) Calculate the percentage of HNCO molecules that are ionised in $0.120 \text{ mol dm}^{-3}$ HNCO .

percentage ionisation of $\text{HNCO} = \dots\dots\dots$ [1]

(e) Silver sulfite, $\text{Ag}_2\text{SO}_3(\text{s})$, is sparingly soluble in water.

(i) Give an expression for the solubility product, K_{sp} , of Ag_2SO_3 .

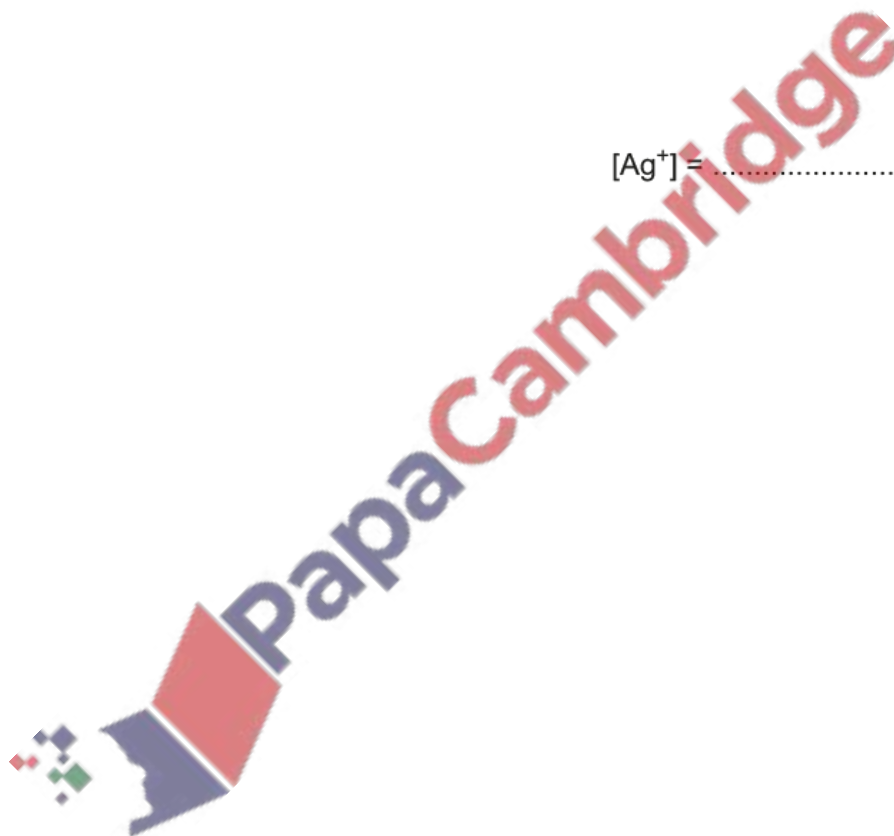
$$K_{\text{sp}} =$$

[1]

(ii) Calculate the equilibrium concentration of Ag^+ in a saturated solution of Ag_2SO_3 at 298 K.

[K_{sp} : Ag_2SO_3 , $1.50 \times 10^{-14} \text{ mol}^3 \text{ dm}^{-9}$ at 298 K]

[Ag^+] = mol dm^{-3} [1]



5. March/2023/Paper_9701/42/No.6(d)

A student uses thin-layer chromatography (TLC) to analyse a mixture containing different metal cations. The student repeats the experiment using different solvents.

Fig. 6.1 shows the chromatogram obtained by the student using water as a solvent.

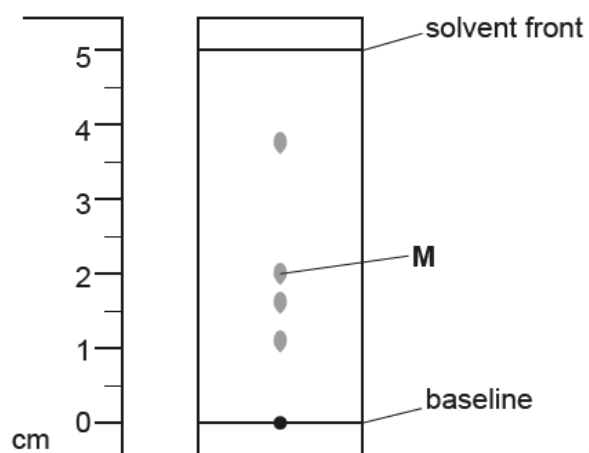


Fig. 6.1

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- (d) In a third experiment, the pH of the mixture of metal ions is kept constant using a buffer solution.

The student prepares the buffer solution by mixing 20.0 cm^3 of 0.150 mol dm^{-3} KOH(aq) and 50.0 cm^3 of 0.100 mol dm^{-3} $\text{C}_8\text{H}_5\text{O}_4\text{K(aq)}$.

$\text{C}_8\text{H}_5\text{O}_4\text{K}$ is a weak carboxylic acid that has $\text{p}K_{\text{a}} = 5.40$.

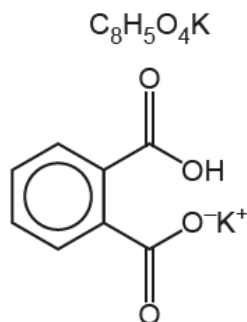


Fig. 6.2

- (i) Complete the equation for the reaction of $\text{C}_8\text{H}_5\text{O}_4\text{K(aq)}$ with KOH(aq) .

$\text{C}_8\text{H}_5\text{O}_4\text{K} + \dots\dots\dots$ [1]

- (ii) Calculate the pH of the buffer solution. Show all your working.

pH = $\dots\dots\dots$ [4]